Sheets Lab Sheets

Sessions 1 to 4 and 24. Work through the tasks in the lab half of the session. Write each answer in the space beside the task.

Sessions 1 to 4 use the workbook lakeside_for_sheets.xlsx. Upload it to Google Drive, open it, and choose File, Save as Google Sheets (in Excel just open it). Session 24 uses a new Google Sheet that you make yourself. Write formulas exactly as the task shows them, with the tab name, an exclamation mark and the range.

The workbook: lakeside_for_sheets.xlsx

Tab	Records	Variables
orders	2,000	order_id, order_date, branch_id, customer_id, payment_method, promo_week, order_value_ugx
order_lines	6,590	line_id, order_id, product_id, quantity, unit_price_ugx, discount, line_revenue_ugx
branches	8	branch_id, branch_name, region, size, opened_year
products	150	product_id, product_name, category, unit_cost_ugx, unit_price_ugx
customers	2,500	customer_id, home_branch_id, signup_date, age_band, loyalty_tier
branch_weeks	832	branch_id, week_start, footfall, rainfall_mm, promo_on, orders, sales_ugx
READ ME	a note	note

Session 1: What data science is

Data file: lakeside_for_sheets.xlsx. Tabs used: orders, order_lines, branches, products, customers, branch_weeks. New tabs you create are named in the tasks.

Part one is in Google Sheets. Part two is the Session 1 notebook in Google Colab.

No	Task	Your answer
1	Open lakeside_for_sheets.xlsx in Google Sheets (File, Import, Upload), or open it in Excel. How many tabs does it have, and what are their names?	
2	Orders tab. Count the records with =COUNTA(orders!A2:A2001) and the variables with =COUNTA(orders!A1:G1). Leave out the header row when you count records.	
3	Order_lines tab. Count its records (=COUNTA(order_lines!A2:A6591)) and variables (=COUNTA(order_lines!A1:G1)). Then divide the lines by the orders: =COUNTA(order_lines!A2:A6591)/COUNTA(orders!A2:A2001).	
4	Write one sentence for each of the two tabs: what does one record stand for?	
5	Find the number of records and variables of branches, products, customers and branch_weeks (the same two formulas, with the tab name and the last row changed).	
6	On the orders tab, count the blank customer_id cells: =COUNTBLANK(orders!D2:D2001). What share of the orders is that, and what do you think the blank means?	
7	An assistant says that about half of the orders are paid by Mobile money. Check it: =COUNTIF(orders!E2:E2001,"Mobile money")/COUNTA(orders!A2:A2001). Format the cell as a percentage.	

No	Task	Your answer
8	Write one business question from your own work, with the decision it would feed and the data you would need. Choose which CRISP-DM phase you think would be hardest.	

Session 2: Data as tables

Data file: lakeside_for_sheets.xlsx. Tabs used: orders, order_lines, branches, products, branch_weeks. New tabs you create are named in the tasks.

No	Task	Your answer
1	Create a new tab called dictionary. In columns A to D list the six data tabs, write one sentence for each saying what a record is, and count its records with =COUNTA(orders!A2:A2001) (change the tab name and last row for each tab).	
2	Orders tab. Predict whether order_id is a primary key. Then test it: =COUNTA(orders!A2:A2001) and =COUNTA(UNIQUE(orders!A2:A2001)).	
3	Order_lines tab. Write what a record is, name its primary key and its two foreign keys. How many different orders do the lines cover (=COUNTA(UNIQUE(order_lines!B2:B6591)))? How many lines does order O000003 have (=COUNTIF(order_lines!B2:B6591,"O000003"))?	
4	Foreign key check. Does every branch_id in orders exist in branches? Use =SUMPRODUCT(--ISNA(MATCH(orders!C2:C2001,branches!A2:A9,0))). Then repeat for product_id in order_lines against products.	
5	Branch_weeks tab. No single variable is unique. Count the different branch_id values and week_start values, multiply the two counts, and compare with the record count. Then test the pair: =COUNTA(UNIQUE(ARRAYFORMULA(branch_weeks!A2:A833&"|"&branch_weeks!B2:B833))).	
6	Is customer_id a primary key of orders? Count the blanks (=COUNTBLANK(orders!D2:D2001)), the filled cells (=COUNTA(orders!D2:D2001)) and the different filled values (=COUNTUNIQUE(orders!D2:D2001)). What share of orders is blank?	
7	Add a variable type column to your dictionary for the seven variables of orders. Then test order_date with =ISNUMBER(orders!B2), =ISTEXT(orders!B2) and =DATEVALUE(orders!B2).	
8	Follow order O000003. How many lines does it have, and what is their total (=SUMIF(order_lines!B2:B6591,"O000003",order_lines!G2:G6591))? Look up its products with =VLOOKUP("P117",products!A2:E151,2,FALSE) and its branch with =VLOOKUP("B01",branches!A2:E9,2,FALSE).	

No	Task	Your answer
9	Categorical variable. Count the orders for each payment method with =COUNTIF(orders!E2:E2001,"Cash"), then Mobile money and Card. How many different values does the variable have (=COUNTA(UNIQUE(orders! E2:E2001)))? Which spelling problems would you look for?	
10	Extreme values. Find the largest order_value_ugx with =MAX(orders!G2:G2001) and the median with =MEDIAN(orders!G2:G2001). Which order is it, at which branch? How many times larger than the median is it? Finish by writing one question you would send to the owner of this data.	

Session 3: Summarising and showing data

Data file: lakeside_for_sheets.xlsx. Tabs used: orders, branches. New tabs you create are named in the tasks.

No	Task	Your answer
1	Open lakeside_for_sheets.xlsx and go to the orders tab. In a new tab called notes, compute for order_value_ugx (column G): the count, the sum, the mean, the median, the mode, the minimum and the maximum. Use COUNT, SUM, AVERAGE, MEDIAN, MODE, MIN and MAX on orders! G2:G2001. Predict first: will the mean be above or below the median?	
2	Compute the quartiles and spread: =QUARTILE(orders! G2:G2001,1), =QUARTILE(orders!G2:G2001,3), the IQR (Q3 minus Q1), the fence (Q3 + 1.5 × IQR), the sample standard deviation =STDEV(orders!G2:G2001), and the number of orders above the fence with =COUNTIF(orders! G2:G2001,">107925").	
3	Draw a histogram of column G: select the column, Insert, Chart, choose Histogram chart. In Customize set the bucket size to 10000 and the outlier percentile to 95. Then check the bars with =COUNTIFS(orders! G2:G2001,"<=10000"), =COUNTIFS(orders! G2:G2001,"<=50000") and =COUNTIF(orders! G2:G2001,">100000"). Describe the shape and name its skewness.	
4	In orders!H1 type month. In H2 type =LEFT(B2,7) and fill down to row 2001. How many orders were placed in December 2024? Use =COUNTIF(H2:H2001,"2024-12").	
5	Build a pivot table: Insert, Pivot table. Rows: branch_id. Columns: month. Values: order_value_ugx summarised by SUM. Read off B01 in 2024-01, B04 in 2024-03, B07 in 2024-01, the 2024-12 column total and the grand total. Predict which cell is unusually large.	
6	Make a second pivot table: Rows: branch_id. Values: order_value_ugx summarised by MEDIAN. Insert a column chart of it and set the vertical axis minimum to 20000. Which branch looks tallest, and how many times taller than the shortest does it look? Then fix the chart: set the minimum to 0 and add a title and axis labels with units.	
7	Make a third pivot table: Rows: month. Values: SUM of order_value_ugx. Insert a line chart with the vertical axis starting at 0. Which month is highest and which is lowest? Does December 2024 stand out?	

No	Task	Your answer
8	Write two sentences for a manager on your notes tab. One gives the typical order value and says how many orders you counted. The other says whether the branches differ much in their typical order.	

Session 4: Uncertainty, samples and cause

Data file: lakeside_for_sheets.xlsx. Tabs used: orders, branch_weeks. New tabs you create are named in the tasks.

No	Task	Your answer
1	Population. On the orders tab, find the mean of all 2,000 order values with =AVERAGE(orders!G2:G2001) and the share of orders above 100,000 UGX with =COUNTIF(orders!G2:G2001,">100000")/COUNT(orders!G2:G2001). Compare the mean with the full-file mean of 47,491 UGX. How big is the gap, and what do we call it?	
2	Simulate. Add a tab called simulation. In A2 put =INDEX(orders!$G$2:$G$2001,1+INT(RAND()*2000)). Fill right to AD2. In AE2 put =AVERAGE(A2:AD2). Select A2:AE2 and fill down to row 201. Copy the block, then Paste special, Values only. Draw a histogram of column AE. Predict first: will the 200 means be spread out or clustered?	
3	Read the spread. In spare cells find the AVERAGE, STDEV, MIN and MAX of column AE. Compare the average with the population mean, and the STDEV with =STDEV.P(orders!G2:G2001)/SQRT(30). What is the STDEV of the 200 sample means called?	
4	Skew. Use =COUNTIF(AE2:AE201,"<"&AVERAGE(orders!G2:G2001)) to count how many of your 200 sample means fall below the population mean. Predict: about half, or more than half? Say why.	
5	Sample size. In AF2 put =AVERAGE(A2:E2) (the first 5 orders only) and fill down to row 201. Copy the column and Paste special, Values only. Find the STDEV of column AF and divide it by the STDEV of column AE. Predict first: bigger or smaller than 1?	
6	Correlation. On the branch_weeks tab find =CORREL(C2:C833,F2:F833) for footfall and orders and =CORREL(D2:D833,F2:F833) for rainfall and orders. Then repeat both for Kampala Central only (B01, rows 2 to 105). Predict whether the B01 correlation is higher or lower.	
7	Selection bias. Compare loyalty orders and walk-in orders: =AVERAGEIF(orders!D2:D2001,"<>",orders!G2:G2001) and =AVERAGEIF(orders!D2:D2001,"",orders!G2:G2001) for the means, =COUNTIF(orders!D2:D2001,"<>") and =COUNTBLANK(orders!D2:D2001) for the counts. Which bias would you introduce by reporting only loyalty orders, and does it matter here?	

No	Task	Your answer
8	Confounder. On branch_weeks use =AVERAGEIF(A2:A833,"B01",C2:C833) and =AVERAGEIF(A2:A833,"B01",F2:F833) (and the same for B07) for mean footfall and orders. A manager sees r = 0.84 and wants flyers to raise footfall. Write three sentences: name a confounder, say what the inside-one-branch figures tell you, and suggest how to test the idea.	

Session 24: Course project and your path

Data file: none. Work in a new Google Sheet named Presentation plan. The tab names are in the tasks.

No	Task	Your answer
1	Open a new Google Sheet and call it Presentation plan. On a tab named plan, list the five slides (Question, Data, Evidence, Risk, Decision) in A2:A6 with these minutes in B2:B6: 1, 2, 3, 2, 2. In B7 add them up. Then write one sentence for each slide about your own course project in C2:C6.	
2	Rehearsal tab. In A2:A6 write the slide names, in B2:B6 the planned seconds (60, 120, 180, 120, 120) and in C2:C6 these illustrative rehearsal seconds: 70, 150, 240, 150, 100. In D2:D6 compute the seconds over plan (=C2-B2). Find the total rehearsal time and the slide that ran longest past its plan.	
3	Evidence tab. A model at decision threshold 0.2 was tested on 2,000 loans with 251 defaults. It flagged 278 loans, and 108 of them really defaulted. Put TP = 108, FP = 170, FN = 143, TN = 1,579 in cells and compute recall, precision and accuracy with formulas. Write one evidence sentence that includes the baseline model (always predicts repaid).	
4	Questions tab. Write three questions your manager or a colleague could ask you about your course project: one on method, one on the metric and one on risk. Write a two-sentence answer to each. Fill with gold the answer that says something is not finished.	
5	Feedback tab. Present your talk to a partner for ten minutes, then swap. For each of the four checklist rows write Yes or Not yet for your partner, and one comment. To practise the formulas, type these illustrative entries in B2:B5: Yes, Not yet, Yes, Yes. Count the rows that still say Not yet and find the first one. Nothing is scored.	
6	Path tab. Write your six month path: three blocks (months 1 and 2, 3 and 4, 5 and 6), one named free course or project in each, and a weekly hour budget of 6 hours. Compute the total hours over 26 weeks and compare with the 48 hours of this course. Rewrite your business question from Session 1 in one sentence using what you know now.	